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IMPERIAL COLLEGE LONDON

DEPARTMENT OF MATHEMATICS

FINAL YEAR RESEARCH PROJECT M3R

Variational Analysis of the Euler Equation on the Rotating Sphere–Energy Structure, Symmetry Breaking and Orbital Stability

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Submitted in partial fulfillment of the requirements for the BSc Mathematics at Imperial College London

Jan 12, 2026

Abstract

Type your abstract here. The abstract is a summary of the contents of the project. It should be brief but informative, and should avoid technicalities as far as possible.

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Chapter 1

Introduction

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Module		
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	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

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Subsubsection Example

Note that you can reference chapters, sections, subsections and subsubsections. For example: Subsection 1.1.1!

1.2 Math Example

While math can be written inline like so $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, we often need to write stand-alone equations like so

$$\text{score}(x) = \left(\lambda_m \sum_{i=0}^{|\mathbf{m}|} \log \hat{p}_m(d(x, \mathbf{m}_i) \mid l_i) \right) + \left(\lambda_l \sum_{i=0}^{|\mathbf{l}|} \log \hat{p}_l(d(x, \mathbf{l}_i) \mid \mathbf{v}_i) \right) + \lambda_p \hat{p}_p(x) \tag{1.1}$$

To write equations over multiple lines (like systems of equation or equations too long to fit the page), one can use the *align* environment coupled to the *subequations* environment like so

$$\mathbb{P}(0 \rightsquigarrow -a) = D \int_0^\infty dt\, k_r e^{-k_r t} \int_0^t d\tau\, \partial_x u\big|_{x=-a}, \tag{1.2a}$$

$$\mathbb{P}(0 \rightsquigarrow b) = -D \int_0^\infty dt\, k_r e^{-k_r t} \int_0^t d\tau\, \partial_x u\big|_{x=b}. \tag{1.2b}$$

1.3 Algorithm Example

See Algorithm 1

Algorithm 1: Algorithm example

Input: $\mathbf{m}$, such that $\mathbf{m}_i$ is the position of the i'th monitor
$\mathbf{l}$, such that $\mathbf{l}_i$ is the position of the i'th landmark
$\mathbf{p}^m$, such that $\mathbf{p}_i^m$ is the ping latency from monitor i to the target
$\mathbf{p}^l$, such that $\mathbf{p}_i^l$ is the set of ping latencies to landmark i

Pre: Compute $\hat{p}_m(d \mid l)$, an estimator giving the likelihood of the target being distance d away from the monitor, given that the monitor records a latency of l to that target. Implemented by training a KDE using $\mathbf{p}^l$.
Compute $\hat{p}_l(d \mid v)$, an estimator giving the likelihood of the target being distance d away from the landmark, given a Canberra distance of v between the target and the landmark, using training targets.

Output: Most likely location of the target

```
Function Likelihood(x, v)
    MonitorScore ← Σ_{i=0}^{|m|} log p̂_m(d(x, m_i) | l_i);
    LandmarkScore ← Σ_{i=0}^{|l|} log p̂_l(d(x, l_i) | v_i);
    return MonitorScore + LandmarkScore
end
v ←{canberra_distance(l_i, p^m) | l_i ∈ l}
C ← Constraint-Based-Geolocation(m, p^m);
C_l ← {m ∈ m | C contains m} ∪ {l ∈ l | C contains l};
return argmax_{x∈C_l} Likelihood(x)
```

1.4 Reference Example

Here is how you can cite papers which you have added in the `/bibs/bibliography.bib` file. You can cite single references as such [1] or multiple references like so [1, 2]. Here is a reference to a website [3].

Chapter 2

Background

2.0.1 Euler's Equation on a rotating sphere

We consider the motion of an ideal incompressible fluid on a rotating sphere S^2 of unit radius. Let (φ, θ) denote the longitude and latitude coordinates, where $\varphi \in [0, 2\pi)$ and $\theta \in [-\pi/2, \pi/2]$. The sphere rotates with a constant angular velocity $\Omega > 0$ around the polar axis.

The velocity field $\mathbf{u}$ is determined by a scalar stream function $\psi : S^2 \to \mathbb{R}$. Specifically, the velocity is given by the rotated gradient of the stream function,

$$\mathbf{u} = \nabla^{\perp}\psi$$

The scalar vorticity ω, defined as the curl of the velocity, is related to the stream function via the Laplace-Beltrami operator on the sphere:

$$\omega = \Delta\psi$$

The dynamics of the fluid are governed by the Euler equations:

$$\partial_t \Delta\psi + \frac{1}{\cos\theta}\left[-\partial_\theta\psi\,\partial_\varphi + \partial_\varphi\psi\,\partial_\theta\right](\Delta\psi + 2\Omega\sin\theta) = 0$$

Due to the rotation of the sphere, the conserved quantity along fluid trajectories is not the relative vorticity ω, but the **absolute vorticity** q, defined as:

$$q = \omega + f(\theta), \quad \text{with } f(\theta) = 2\Omega\sin\theta$$

where $f(\theta)$ is the Coriolis parameter. We recall the velocity field version of Euler's Equation is given by:

$$\begin{cases} D_t\mathbf{u} + 2\Omega\sin\theta\, J\mathbf{u} = \partial_t\mathbf{u} + \nabla_{\mathbf{u}}\mathbf{u} + +2\Omega\sin\theta\, J\mathbf{u} = -\nabla p, \\ \operatorname{div}\mathbf{u} = 0 \end{cases}$$

here the term ∇p arises as a Lagrange Multiplier from the divergence-free constraint. Alternatively, the equation at the level of vorticity is given by the transport equation:

$$\partial_t q + \mathbf{u}\cdot\nabla q = 0$$

or equivalently in terms of material derivatives:

$$D_t q = D_t(\omega + 2\Omega\sin\theta) = 0$$

2.0.2 Conservation Laws and Functional Setting

We will work in the zero-mean subspace (to satisfy the compatibility condition):

$$L_0^2(S^2) = \{w : \int_{S^2} w \, d\sigma = 0\}$$

due to a single consequence from Divergence Theorem, note the sphere has zero boundaries:

$$\int_{S^2} \Delta\psi \, d\sigma = \int_{S^2} \text{div}(\nabla\psi) \, d\sigma = \int_{\partial S^2} \nabla\psi \cdot \mathbf{n} \, ds = 0$$

We denote by $d\sigma$ the Riemannian area element on S^2, where this reads $d\sigma = \cos\theta \, d\varphi \, d\theta$ in longitude–latitude coordinates

$$(\varphi, \theta) \in [0, 2\pi) \times (-\pi/2, \pi/2)$$

Energy defined by:

$$E(t) = \frac{1}{2} \int_{S^2} |\mathbf{u}|^2 \, d\sigma$$

is conserved since we have:

$$\frac{dE}{dt} = -\int_{S^2} (\nabla_{\mathbf{u}} \mathbf{u}) \cdot \mathbf{u} \, d\sigma - \int_{S^2} f(\theta)(J\mathbf{u}) \cdot \mathbf{u} \, d\sigma - \int_{S^2} (\nabla p) \cdot \mathbf{u} \, d\sigma = 0$$

because $(J\mathbf{u}) \cdot \mathbf{u} = 0$, and $\int (\nabla p) \cdot \mathbf{u} = -\int p \, \text{div} \, \mathbf{u} = 0$, as well as $(\nabla_{\mathbf{u}} \mathbf{u}) \cdot \mathbf{u} = \frac{1}{2}\mathbf{u}(|\mathbf{u}|^2) = \text{div}\left(\frac{1}{2}|\mathbf{u}|^2\mathbf{u}\right)$ integrates to 0 (the sphere has no boundary).

Casimir Invariants is conserved, see appendix.

Angular Momentum is conserved.

Chapter 3

Variation Methods and Stability Analysis

3.1 Variational Principle on the Rotating Sphere

Following the approach of Coti Zelati and Delgadino [2405], we propose to predict the long-time behavior of the fluid by maximizing the mixing entropy subject to the conservation of energy.

3.1.1 Revisiting the Min–Max Principle in 2405

We recall the variational formulation introduced in 2405. Given an admissible class $\mathcal{O}_{\text{in}}$, the maximal entropy at fixed energy level e is defined by the microcanonical problem

$$S(e) := \max \{S_f(\omega) : \omega \in \mathcal{O}_{\text{in}},\ E(\omega) = e\} \tag{1.6}$$

The energy constraint can be rewritten using a Lagrange multiplier $\beta \in \mathbb{R}$ as

$$S(e) = \max_{\omega \in \mathcal{O}_{\text{in}}} \left\{ S_f(\omega) + \inf_{\beta \in \mathbb{R}} \beta\big(e - E(\omega)\big) \right\} \tag{1.7}$$

because we encoded the constraint of constant energy $E(w)$ using the punishing term:

$$\inf_{\beta \in \mathbb{R}} \beta(e - E(\omega)) = \begin{cases} 0 & \text{if } E(\omega) = e, \\ -\infty & \text{otherwise} \end{cases}$$

By formally commuting the $\max$ and $\inf$ (justified through the min–max principle), one obtains

$$S(e) = \inf_{\beta \in \mathbb{R}} \left\{ \beta e + \max_{\omega \in \mathcal{O}_{\text{in}}} \big(S_f(\omega) - \beta E(\omega)\big) \right\} \tag{1.8}$$

Introducing the free energy functional $F_\beta(\omega) := S_f(\omega) - \beta E(\omega)$ and defining $g(\beta) := \max_{\omega \in \mathcal{O}_{\text{in}}} F_\beta(\omega)$ we can rewrite

$$S(e) = \inf_{\beta \in \mathbb{R}} \{g(\beta) + \beta e\}$$

A fundamental inequality giving upper bound for the constraint Entropy For any $\beta \in \mathbb{R}$ and any $\omega \in \mathcal{O}_{\text{in}}$, we have $F_\beta(\omega) = S_f(\omega) - \beta E(\omega)$, hence

$$S_f(\omega) = F_\beta(\omega) + \beta E(\omega)$$

By definition of $g(\beta)$, we have $F_\beta(\omega) \leq g(\beta)$ and therefore $S_f(\omega) \leq g(\beta) + \beta E(\omega)$. Restricting to configurations satisfying the energy constraint $E(\omega) = e$, we obtain

$$S_f(\omega) \leq g(\beta) + \beta e$$

Taking the supremum over all such ω satisfying energy constraints, we deduce

$$S(e) \leq g(\beta) + \beta e, \quad \forall \beta \in \mathbb{R} \tag{3.1}$$

Consequently we could take all infimum over β and arrive at:

$$S(e) \leq \inf_{\beta \in \mathbb{R}} \{g(\beta) + \beta e\} \tag{3.2}$$

Sharpness via the min–max principle. The min–max principle asserts that the above inequality is in fact sharp, namely

$$S(e) = \inf_{\beta \in \mathbb{R}} \{g(\beta) + \beta e\} \tag{3.3}$$

Let ω_β be a maximizer of F_β under a fixed β in the inner part, where we maximize over the free energy F_β on the orbit $\mathcal{O}_{\text{in}}$, and then let β^* be a minimizer of the whole right-hand side in the outer layer, then

$$g(\beta^*) = F_{\beta^*}(\omega_{\beta^*}) = S_f(\omega_{\beta^*}) - \beta^* E(\omega_{\beta^*})$$

and therefore

$$S(e) = S_f(\omega_{\beta^*}) + \beta^* \big(e - E(\omega_{\beta^*})\big) \tag{3.4}$$

To justify rigorously that the free-energy maximizer selected by the outer minimization indeed satisfies the energy constraint and coincides with the entropy maximizer, we follow the subdifferential argument used in 2405. Recall that

$$g(\beta) := \max_{\omega \in \mathcal{O}_{\text{in}}} F_\beta(\omega), \qquad \text{where} \quad F_\beta(\omega) = S_f(\omega) - \beta E(\omega)$$

so that

$$S(e) = \inf_{\beta \in \mathbb{R}} \{g(\beta) + \beta e\}$$

Let

$$\beta(e) := \{\beta \in \mathbb{R} : S(e) = g(\beta) + \beta e\}$$

If $\beta^* \in \beta(e)$, then β^* is a minimizer of the convex function:

$$\beta \mapsto g(\beta) + \beta e$$

Therefore, by the first-order optimality condition for convex functions (a generalization of vanishing condition of derivatives for differentiable functions):

$$0 \in \partial\big(g(\beta) + \beta e\big)\big|_{\beta=\beta^*} = \partial g(\beta^*) + e$$

which is equivalent to

$$-e \in \partial g(\beta^*) \tag{3.5}$$

Next, since g is the pointwise supremum of the affine functions

$$G_\omega(\beta) := S_f(\omega) - \beta E(\omega),$$

the general formula for the subdifferential of a supremum gives

$$\partial g(\beta) = \mathrm{co}\Big(\bigcup_{\omega:\, G_\omega(\beta)=g(\beta)} \partial G_\omega(\beta) \Big)$$

Under the uniqueness assumption for the maximizer of F_β, the active set consists only of ω_β, and since G_ω is differentiable in β, this reduces to

$$\partial g(\beta) = \{-E(\omega_\beta)\} \tag{3.6}$$

Evaluating at $\beta = \beta^*$ and combining previous conclusions that $-e \in \partial g(\beta^*)$, we obtain

$$-e = -E(\omega_{\beta^*}) \quad \Rightarrow \quad e = E(\omega_{\beta^*})$$

Substituting back into (3.4), we finally deduce

$$S(e) = S_f(\omega_{\beta^*})$$

so that ω_{β^*} is not only the maximizer of the free energy F_{β^*}, but also the entropy maximizer under the fixed energy constraint $E(\omega) = e$. In this sense, the min–max principle provides the precise bridge between the canonical and microcanonical variational problems.

3.1.2 Microcanonical Formulation and the Min–Max Principle on Rotating Sphere

Firstly define our rearrangement class (isovortical orbit) as:

$$\mathcal{O}_{q_0} = \{q_0 \circ \Phi^{-1} : \Phi \in \mathrm{SDiff}(S^2)\}$$

We consider the entropy maximization problem at fixed energy:

$$S(e) = \sup\{S(q) : q \in \mathcal{O}_{q_0},\ E(q) = e\}$$

Following the approach of 2405, we encode the energy constraint using a dual parameter $\beta \in \mathbb{R}$:

$$S(e) = \sup_{q \in \mathcal{O}_{q_0}} \left\{ S(q) + \inf_{\beta \in \mathbb{R}} \beta(e - E(q)) \right\}$$

Formally exchanging $\sup$ and $\inf$, we obtain

$$S(e) = \inf_{\beta \in \mathbb{R}} \left\{ \beta e + \sup_{q \in \mathcal{O}_{q_0}} \big(S(q) - \beta E(q) \big) \right\}$$

This motivates the definition of the free energy functional

$$\mathcal{F}_\beta(q) := S(q) - \beta E(q)$$

3.1.3 Free Energy Functional

Let $q = \omega + f(\theta)$ be the absolute vorticity, where $f(\theta) = 2\Omega \sin\theta$. We consider the mixing entropy functional defined on the absolute vorticity:

$$\mathcal{S}(q) = -\int_{S^2} q \log q \, d\sigma. \tag{3.5}$$

We aim to maximize $\mathcal{S}$ while keeping the kinetic energy $E(\omega)$ constant. Motivated by the above min–max formulation, we consider the free energy functional, we define the free energy functional for the vorticity field ω, as inspired by Statistical Mechanics:

$$\mathcal{F}_\beta(\omega) = \mathcal{S}(\omega + f) - \beta E(\omega). \tag{3.6}$$

Note that the geometry of the sphere and the rotation enter the functional through the definition of q.

3.1.4 The Euler-Lagrange Equation

A critical point ω^* of $\mathcal{F}_\beta$ satisfies the first variation condition $\delta\mathcal{F}_\beta(\omega^*) = 0$. Computing the functional derivatives with respect to ω:

$$\frac{\delta \mathcal{S}}{\delta \omega} = -(\log(\omega + f) + 1), \tag{3.7}$$

$$\frac{\delta E}{\delta \omega} = -\psi, \tag{3.8}$$

we obtain the relation:

$$\log(\omega^* + f) + 1 = \beta\psi^* + \text{const} \tag{3.9}$$

Rearranging terms, the entropy maximizer satisfies the semilinear elliptic equation:

$$\Delta\psi^* + 2\Omega \sin\theta = \lambda e^{\beta\psi^*} \tag{3.10}$$

where λ is a normalization constant related to the total circulation/mass.

3.1.5 Connection to Rigidity Results

Equation (3.10) falls precisely into the class of stationary solutions studied by Constantin and Germain [2109], of the form $\Delta\psi + f = F(\psi)$, indicating that a state maximizing entropy must be a steady state. This connection allows us to apply the rigidity and stability criteria from [2109] to analyze the properties of our entropy maximizers.

3.2 Results: Structure and Stability of Maximizers

In this section, we analyze the physical structure of the entropy maximizers derived in Eq. (3.10). We focus on two key aspects: the explicit form of solutions in the high-temperature limit using spherical harmonics, and their dynamical stability based on the criteria from Constantin and Germain [2109].

3.2.1 The Linear Limit and Mode Selection

The nonlinear relation $q = \lambda e^{\beta\psi}$ is generally difficult to solve explicitly. However, in the limit of small inverse temperature $|\beta| \ll 1$ (corresponding to high noise/mixing), we can Taylor expand the exponential term $e^{\beta\psi} \approx 1 + \beta\psi$. The Euler-Lagrange equation (3.10) simplifies to:

$$\Delta\psi - \lambda\beta\psi = -2\Omega \sin\theta + \lambda \tag{3.11}$$

This is a linear Helmholtz-type equation on the sphere. To solve it, we expand the stream function in the basis of spherical harmonics $Y_{l,m}(\theta, \varphi)$, which are eigenfunctions of the Laplace-Beltrami operator:

$$\Delta Y_{l,m} = -l(l+1)Y_{l,m}, \quad l \geq 0,\ |m| \leq l \tag{3.12}$$

Now we first rearrange (3.11) and substitute in the eigenfunction decomposition of ψ gives:

$$(\Delta - \lambda\beta)\psi = \sum_{l,m} a_{l,m}(\Delta - \lambda\beta)Y_{l,m} = \sum_{l,m} a_{l,m}(-l(l+1) - \lambda\beta)Y_{l,m}$$

while the right hand side becomes:

$$C(\theta, \varphi) = \sum_{l,m} c_{l,m} Y_{l,m}(\theta, \varphi), \quad c_{l,m} = \int_{S^2} C\,\overline{Y_{l,m}}\, d\sigma$$

Comparing coefficients gives:

$$a_{l,m} = \frac{c_{l,m}}{-l(l+1) - \lambda\beta}$$

Noting that the Coriolis term $2\Omega \sin\theta$ projects onto the $l = 1$ zonal mode (specifically proportional to $Y_{1,0}$) and that the constant λ corresponds to $(l, m) = (0, 0)$. Eq. (3.11) suggests that the entropy maximizer ψ is dominated by low-order modes. We see in this case our steady solution is zonal.

3.2.2 Dynamical Stability Analysis

We now discuss whether the stationary states corresponding to entropy maximizer are dynamically stable for the Euler's equation on the rotating sphere. Throughout, we use the concept of *Lyapunov stability in* $H^2(S^2)$, as in [2109].
Recall that our entropy maximizers satisfy the steady equation:

$$\Delta\psi + 2\Omega \sin\theta = F(\psi), \quad F(\psi) = \lambda e^{\beta\psi} \tag{3.13}$$

In particular,

$$F'(\psi) = \beta\lambda e^{\beta\psi} = \beta\, F(\psi) \tag{3.14}$$

Arnold-type nonlinear stability. A key fact used in [2109] is that on the sphere, the relevant stability threshold is the second eigenvalue -6 of the Laplace-Beltrami Operator Δ.
If $\beta > 0$ then $F'(\psi) > 0$, we conclude from classical Arnold Theory that solution is Arnold stable of type II. Hence all entropy maximizers under positive temperature is nonlinearly stable.

If $\beta < 0$ and a stationary solution of (3.13) satisfies

$$-6 < F'(\psi) < 0 \tag{3.15}$$

then it is Lyapunov stable in $H^2(S^2)$ according to Theorem 2 in [2109]. We again note that by referencing Theorem 4 (Rigidity Analysis) in [2109], we get all solutions ψ are Zonal whenever $F'(\psi) > -6$. Hence all entropy maximizers prescribe rotational symmetry. As a result, genuine non-zonal structures may only start to appear when $F'(\psi) \approx -6$, where Rossby–Haurwitz solutions of degree 2 reside. To rigorously understand the onset of non-zonal instabilities, we perform a linear perturbation analysis around a stationary zonal solution ψ_0. Suppose ψ_0 satisfies the steady Euler equation:

$$\Delta\psi_0 + 2\Omega \sin\theta = F(\psi_0) \tag{3.16}$$

We introduce a small perturbation $\epsilon\phi(\theta, \varphi)$ such that the perturbed state is $\psi = \psi_0 + \epsilon\phi$. Substituting this into the governing equation and performing a Taylor expansion on the nonlinear term F yields:

$$\Delta(\psi_0 + \epsilon\phi) + 2\Omega \sin\theta = F(\psi_0) + \epsilon F'(\psi_0)\phi + \mathcal{O}(\epsilon^2) \tag{3.17}$$

Subtracting the base state equation for ψ_0 and dividing by ϵ, we obtain the linearized eigenvalue problem for the perturbation ϕ at the leading order:

$$\Delta\phi = F'(\psi_0)\phi \tag{3.18}$$

This reveals that the perturbation ϕ must be an eigenfunction of the Laplace-Beltrami operator Δ_{S^2}, which acts on the spherical harmonics $Y_{l,m}$ with eigenvalues $-l(l+1)$. Theorem 4 in [2109] enforces rigidity (and hence strictly zonal flows) whenever $F'(\psi) > -6$.

Therefore, a symmetry-breaking bifurcation can only manifest when the effective parameter $F'(\psi_0)$ reaches the critical threshold of -6. At this juncture, the perturbation ϕ is permitted to reside in the eigenspace E_2 (where $l = 2$). Since E_2 contains modes with $m \neq 0$, the activation of these modes breaks the continuous longitudinal symmetry of the zonal flow, driving the system into a non-zonal Rossby-Haurwitz wave regime.
When $F'(\psi) = -6$, solution lies in $E_1 + E_2$ (rotation + RH waves) and is of the form:

$$\psi_0 = \alpha \sin\theta + Y_2, \quad Y_2 \in E_2$$

and Theorem 3 in [2109] says that $E_1 + E_2$ is stable in $H^2(S^2)$. And when $F'(\psi)$ continues to decrease, the non-zonal RH waves loses stability while zonal RH waves maintain stability.

3.2.3 Physical Implications

The combination of entropy maximization and rigidity results leads to a strong physical conclusion:

1. Entropy maximization drives the system towards a state $q = \lambda e^{\beta\psi}$.
2. Rigidity theorems (Theorem 4 in [2109]) imply that for a wide range of parameters (especially when $F' > -6$), these solutions must be **Zonal Flows**.

This suggests that the "stripes" seen on Jupiter are not just dynamically stable, but are the statistically most probable state of a turbulent fluid on a rotating sphere.

Chapter 4

Energy Maximization and Characterization of Dipole States

4.1 Symmetry Breaking and Energy Upper Bound for Zonal Flows

Recall that the free energy functional is defined by

$$\mathcal{F}_\beta = S - \beta E$$

The sign of β determines which mechanism dominates the variational principle.

Positive temperature regime ($\beta > 0$) In this case, $-\beta E$ is negative, and maximizing $\mathcal{F}_\beta$ favors increasing the entropy S while penalizing large energy. The system therefore tends to homogenize the vorticity distribution. On the sphere, this corresponds to the zonal flows obtained in Chapter 3, which represent maximally mixed states under the energy constraint.

Negative temperature regime ($\beta < 0$) When the prescribed energy exceeds the maximal energy admissible by zonal states (denoted $E^{\max}_{\text{zonal}}$), no positive temperature equilibrium exists. The Lagrange multiplier β is then forced to become negative and $\mathcal{F}_\beta = S + |\beta|E$ so that energy is rewarded rather than penalized.

In this section, we investigate the behavior of the system in the high-energy (large $\beta < 0$) regime. By drawing an analogy to the radial symmetry constraints on a 2D planar domain explored in [2405], we impose a strict zonal symmetry constraint on the sphere. Zonal symmetry implies that the fluid state depends only on the latitude θ and is independent of the longitude φ. In the context of spherical harmonics, this restricts our solutions to the $m = 0$ modes. We demonstrate that under the strict assumption of zonal symmetry, the total kinetic energy of the fluid is strictly bounded from above, given constant enstrophy and angular momentum.

4.1.1 Derivation of the Zonal Energy Bound

Let the relative vorticity $\omega \in L^2_0(S^2)$ be strictly zonal. It can be decomposed using the zonal spherical harmonics $Y_{l,0}(\theta)$ as follows:

$$\omega(\theta) = \sum_{l=1}^{\infty} c_{l,0} Y_{l,0}(\theta) \tag{4.1}$$

Note that the $l = 0$ term vanishes due to the zero-mean condition:

$$c_{0,0} = \int_{S^2} \omega \overline{Y_{0,0}}\, d\sigma = \int_{S^2} \omega \left(\frac{1}{\sqrt{4\pi}}\right) d\sigma = \frac{1}{\sqrt{4\pi}} \int_{S^2} \omega\, d\sigma = 0$$

The corresponding stream function ψ, satisfying $\Delta\psi = \omega$, is given by:

$$\psi(\theta) = \sum_{l=1}^{\infty} \frac{-c_{l,0}}{l(l+1)} Y_{l,0}(\theta) \tag{4.2}$$

The kinetic energy $E(\omega) = \frac{1}{2}\int_{S^2} |\mathbf{u}|^2 d\sigma$ can be rewritten in terms of ψ and ω. Since $\mathbf{u} = \nabla^{\perp}\psi$, we have $|\mathbf{u}|^2 = |\nabla\psi|^2$. Integrating by parts over the sphere (which has no boundary) yields:

$$E(\omega) = \frac{1}{2}\int_{S^2} |\nabla\psi|^2 d\sigma = -\frac{1}{2}\int_{S^2} \psi\Delta\psi d\sigma = -\frac{1}{2}\int_{S^2} \psi\omega d\sigma \tag{4.3}$$

Substituting the harmonic expansions and applying the orthonormality of $Y_{l,0}$, we obtain the energy in spectral space:

$$E(\omega) = \frac{1}{2}\sum_{l=1}^{\infty} \frac{|c_{l,0}|^2}{l(l+1)} \tag{4.4}$$

We now introduce two conserved physical quantities:

1. **Angular Momentum:** The angular momentum corresponds to the $l = 1$ projection of the vorticity. Thus, the coefficient $c_{1,0}$ is fixed by the initial conditions, which we denote as a constant A.
2. **Enstrophy (Z):** As a specific case of Casimir invariants $f(q) = q^2$, the enstrophy is a conserved constant Z_0.

Here since we are using absolute vorticity:

$$q(\theta) = \omega(\theta) + f(\theta) = (c_{1,0} + 2\Omega\sqrt{\frac{4\pi}{3}})Y_{1,0}(\theta) + \sum_{l=2}^{\infty} c_{l,0}Y_{l,0}(\theta), \quad \text{since} \quad f(\theta) = 2\Omega\sin\theta = 2\Omega\sqrt{\frac{4\pi}{3}}Y_{1,0}(\theta)$$

Using orthogonality, the enstrophy is then given by the Parseval's Equality:

$$Z_q = \int_{S^2} q^2\, d\sigma = |c_{1,0} + 2\Omega\sqrt{\frac{4\pi}{3}}|^2 + \sum_{l=2}^{\infty} |c_{l,0}|^2$$

Splitting the energy sum into the $l = 1$ mode and the higher-order modes, we have:

$$E(\omega) = \frac{1}{2}\frac{|c_{1,0}|^2}{1(1+1)} + \frac{1}{2}\sum_{l=2}^{\infty} \frac{|c_{l,0}|^2}{l(l+1)} \tag{4.5}$$

Since $l(l+1) \geq 6$ for all $l \geq 2$, we can establish an upper bound for the series:

$$E(\omega) \leq \frac{A^2}{4} + \frac{1}{12}\sum_{l=2}^{\infty} |c_{l,0}|^2 \tag{4.6}$$

Using the enstrophy conservation, we arrive at the strict upper bound for the kinetic energy of any zonal flow:

$$E_{zonal}^{max} \leq \frac{|A|^2}{4} + \frac{1}{12}\left(Z_q - |A + 2\Omega\sqrt{\frac{4\pi}{3}}|^2\right) \tag{4.7}$$

4.2 The Point Vortex Limit and Energy Divergence

As established in the previous section, maximizing the free energy $\mathcal{J}_\beta$ in the large β limit requires breaking the zonal symmetry to surpass the kinetic energy bound E_{zonal}^{max}. We now provide a rigorous mathematical justification for how non-zonal localized structures, specifically point vortices, can accommodate this divergent energy requirement.

Suppose the vorticity field strongly concentrates around a point $x_0 \in S^2$. In the low-temperature limit $\beta \to -\infty$, vorticity may strongly concentrate around isolated points. Formally, such concentration can be described in the sense of Dirac Delta measures as $\omega \to \Gamma\delta_{x_0}$, where the coefficient Γ represents the local circulation of the patch:

$$\Gamma = \int_{C_r(x_0)} \omega \, d\sigma$$

The stream function satisfies the Poisson equation $\Delta\psi = \Gamma\delta_{x_0}$. The solution to this equation is fundamentally the Green's function for the Laplace-Beltrami operator on the sphere.

In the immediate infinitesimal neighborhood of x_0, the geometry of the sphere is locally flat, and the Green's function exhibits the well-known logarithmic singularity characteristic of two-dimensional domains:

$$\psi(r) \sim \frac{\Gamma}{2\pi} \ln r \quad \text{as} \quad r \to 0 \tag{4.8}$$

where r is the geodesic distance from x_0. The magnitude of the velocity field $\mathbf{u} = \nabla^\perp\psi$ diverges as:

$$|\mathbf{u}(r)| = |\nabla\psi| \sim \frac{\Gamma}{2\pi r} \tag{4.9}$$

We now calculate the local contribution of this concentrated vortex core to the total kinetic energy. Integrating over a small disk $B_R(x_0)$ of radius $R \ll 1$ (where the local planar approximation is rigorously valid), the local kinetic energy is:

$$E_{local} = \frac{1}{2}\int_{B_R(x_0)} |\mathbf{u}|^2 d\sigma \sim \frac{1}{2}\int_0^{2\pi}\int_0^R \left(\frac{\Gamma}{2\pi r}\right)^2 r dr d\varphi = \frac{\Gamma^2}{4\pi}\int_0^R \frac{1}{r} dr \tag{4.10}$$

Crucially, the integral $\int_0^R r^{-1} dr = \ln(R) - \ln(r)\big|_{r\to 0}$ diverges to $+\infty$ at the lower limit $r \to 0$. This divergence demonstrates that by concentrating vorticity into an infinitesimally small non-zonal region, the fluid can theoretically contain infinite kinetic energy, thus breaking the zonal upper bound E_{zonal}^{max}.

4.3 Construct Energy Lower Bound for Non-Zonal Vortex Patches

In the previous section, we established the upper bound for the energy of purely zonal flows, denoted as E_{zonal}^{max}. However, as $\beta \to -\infty$, the system tends to maximize entropy by forming localized, intense vortex structures. To rigorously show that such symmetry-breaking occurs, we construct a non-zonal vortex patch and compute its energy lower bound.

Let us define a spherical cap region $C_r(x_0)$ centered at an arbitrary pole $x_0 \in S^2$ with geodesic radius r. In a local spherical coordinate system (ϑ, ϕ) centered at x_0, the region is defined by $0 \le \theta \le r$. The area of this spherical cap is given by:

$$|C_r| = \int_0^{2\pi}\int_0^r \sin\vartheta d\vartheta d\phi = 2\pi(1 - \cos r) \tag{4.11}$$

We consider a uniform vortex patch with a very large, constant absolute vorticity L confined within $C_r(x_0)$, such that $q_L(x) = L\mathbb{I}_{C_r}(x)$. The local absolute circulation Γ is conserved as a result of the transport equation and is given by:

$$\Gamma_q = \int_{C_r} q_L d\sigma = \int_{C_r} L d\sigma = 2\pi L(1 - \cos r) \tag{4.12}$$

The local relative circulation Γ_ω (which is the actual source generating velocity and kinetic energy in the Green's function formulation) is:

$$\Gamma_\omega = \int_{C_r} \omega_L \, d\sigma = \int_{C_r} (L - 2\Omega \sin\theta) \, d\sigma = L|C_r| - \int_{C_r} 2\Omega \sin\theta \, d\sigma$$

The key point is: as $L \to \infty$, in order to keep the circulation Γ_q finite, the area $|C_r| \to 0$. Meanwhile, the absolute value of $2\Omega \sin\theta$ is at most 2Ω. Therefore:

$$\left| \int_{C_r} 2\Omega \sin\theta \, d\sigma \right| \leq 2\Omega |C_r| = \frac{2\Omega \Gamma_q}{L} \to 0$$

Hence we could use Γ_w as an equivalent notion instead of Γ_q to analyze the asymptotic behavior and denote $\Gamma_w \approx \Gamma_q := \Gamma$. For highly concentrated vortices where L is large, the radius r becomes arbitrarily small. Using the Taylor expansion $\cos r \approx 1 - r^2/2$, we obtain the asymptotic relation:

$$r \approx \sqrt{\frac{\Gamma}{\pi L}} \tag{4.13}$$

The kinetic energy of the flow is determined by the Green's function of the Laplace-Beltrami operator on S^2, which has the fundamental singular form $G(x, y) = \frac{1}{4\pi} \ln\left(\frac{|x-y|^2}{2}\right) + C$:

$$E = \frac{1}{2} \int_{S^2} |\mathbf{u}|^2 \, d\sigma = \frac{1}{2} \int_{S^2} \nabla\psi \cdot \nabla\psi \, d\sigma = -\frac{1}{2} \int_{S^2} \psi \Delta\psi \, d\sigma$$

and by substituting the convolution expression of stream function ψ:

$$E = -\frac{1}{2} \int_{S^2} \left(\int_{S^2} G(x, y) \, \omega_L(y) \, d\sigma(y) \right) \omega_L(x) \, d\sigma(x)$$

Up to an $\mathcal{O}(1)$ constant associated with total circulation, the dominant singular part of the energy is:

$$E(\omega_L) \approx -\frac{1}{8\pi} \iint_{C_r \times C_r} \ln\left(\frac{|x-y|^2}{2}\right) \omega_L(x) \omega_L(y) d\sigma(x) d\sigma(y) \tag{4.14}$$

Notice that for any two points $x, y \in C_r$, their Euclidean chord length $|x - y|$ is strictly bounded by the maximum surface diameter of the cap, which is $2r$. Thus, we have the geometric inequality $|x - y| \leq 2r$. This allows us to establish a strict lower bound for the logarithmic kernel:

$$2 \ln\left(\frac{|x-y|}{\sqrt{2}}\right) \leq 2 \ln\left(\frac{2r}{\sqrt{2}}\right) = 2 \ln\left(\sqrt{2} r\right) \tag{4.15}$$

Substituting this lower bound into the double integral yields:

$$\begin{aligned} E(\omega_L) &\geq \frac{1}{4\pi} \ln\left(\frac{1}{\sqrt{2} r}\right) \iint_{C_r \times C_r} L^2 d\sigma(x) d\sigma(y) \\ &= \frac{1}{4\pi} \ln\left(\frac{1}{\sqrt{2} r}\right) \Gamma^2 \end{aligned} \tag{4.16}$$

Using the asymptotic approximation $r \approx \sqrt{\Gamma/(\pi L)}$, we obtain the final lower bound for the energy of the localized vortex patch:

$$E(\omega_L) \geq \frac{\Gamma^2}{8\pi} \ln\left(\frac{\pi L}{2\Gamma}\right) \tag{4.17}$$

Conclusion: As the local maximum vorticity $L \to \infty$ (corresponding to the large β limit), the energy of the localized vortex patch diverges logarithmically, i.e., $E(\omega_L) \propto \Gamma^2 \ln L \to +\infty$. Consequently, for a sufficiently large L, this energy strictly exceeds the finite upper bound E_{zonal}^{max}. This analytically demonstrates that the absolute energy maximum on the rotating sphere cannot remain purely zonal; the system must undergo symmetry breaking, leading to the emergence of localized point vortices or dipoles.

4.4 Asymptotic Concentration for Large Negative $\beta < 0$

The physical intuition of point vortex formation is independently corroborated by the asymptotic behavior of the Euler-Lagrange equation (Eq. 3.10). Recall the steady state equation for the absolute vorticity $q = \lambda e^{\beta\psi}$.

Let x_0 be the point on the sphere where the stream function ψ attains its global minimum, $\psi_{min} = \psi(x_0)$. We examine the absolute vorticity at any other point x compared to the maximum:

$$\frac{q(x)}{q(x_0)} = \frac{\lambda e^{\beta\psi(x)}}{\lambda e^{\beta\psi_{min}}} = e^{\beta(\psi(x)-\psi_{min})} \tag{4.18}$$

Since ψ_{max} is the global maximum, the exponent $\psi(x) - \psi_{min}$ is strictly positive for all $x \neq x_0$. Therefore, in the low-temperature limit ($\beta \to -\infty$):

$$\lim_{\beta\to-\infty} e^{\beta(\psi(x)-\psi_{min})} = 0 \quad \text{for all} \quad x \neq x_0 \tag{4.19}$$

This asymptotic limit implies that the absolute vorticity vanishes exponentially fast everywhere except at the exact location of the stream function's minimum. Mathematically, the continuous vorticity distribution q_{abs} converges to a sum of Dirac delta measures. This again proves that the statistical mechanics formulation inherently forces the fluid into a point vortex state when seeking maximum energy.

4.5 Emergence of Dipole from Existence of Point Vortex

However, on the sphere the vorticity satisfies the structural constraint

$$\int_{S^2} \omega \, d\sigma = 0$$

which follows from the identity $\omega = \Delta\psi$ and the absence of boundary. Therefore, a single concentrated vortex is not admissible unless $\Gamma = 0$. The minimal admissible singular configuration is thus a dipole given by:

$$\omega = \Gamma\delta_{x_+} - \Gamma\delta_{x_-}$$

consisting of two point vortices with equal magnitude and opposite circulation. Consequently, the symmetry-breaking phase transition does not merely yield isolated spots, but fundamentally forces the emergence of robust, self-propelling vortex pairs known as **Dipoles** (or spherical Modons). Details of Self Propelling could be found at Appendix section.

4.5.1 Physical Implications and the Role of Enstrophy

In the context of 2D and geophysical fluid dynamics, an isovortical orbit $\mathcal{O}_{q_0}$ is defined as the set of all vorticity fields reachable from an initial state q_0 via area-preserving diffeomorphisms, mathematically represented as $\mathcal{O}_{q_0} = \{q_0 \circ \Phi^{-1} \mid \Phi \in \text{SDiff}(S^2)\}$. By definition, any functional of the form $C[q] = \int_{S^2} f(q) d\sigma$ constitutes a Casimir invariant and is strictly conserved along this orbit.
Instead of undertaking the complex functional analysis required to characterize the absolute maximum energy over the entire geometrical orbit $\mathcal{O}_{\omega_0}$ as in [2405], we isolate a single, specific Casimir invariant: the enstrophy $Z = \int_{S^2} q^2 d\sigma$. Because enstrophy is conserved on the orbit, leveraging it as a uniform L^2 bound is entirely sufficient to rigorously establish the kinetic energy ceiling E_{zonal}^{max} for any state that strictly maintains $m = 0$ continuous longitudinal symmetry.

This upper bound reveals a profound phase transition mechanism in the low-temperature (large β) limit. When the free energy functional $\mathcal{J}_\beta$ aggressively demands the maximization of kinetic energy, the system may require an energy level that strictly exceeds E_{zonal}^{max}. When this threshold is crossed, the fluid is mathematically compelled to break the zonal symmetry. Physically, this symmetry breaking allows vorticity to pool and concentrate into localized, highly intense structures, such as coherent dipoles or point-like vortices. Because kinetic energy scales with the squared gradient of the stream function ($E = \frac{1}{2} \int_{S^2} |\nabla \psi|^2 d\sigma$), these highly concentrated non-zonal structures can contain significantly more kinetic energy than uniformly distributed zonal bands, providing a clear physical mechanism for the bifurcation of atmospheric stripes into isolated storms.

Chapter 5

Stability of Emergent Dipole

In Chapter 4, we rigorously demonstrated that in the regime of large negative inverse temperature ($\beta \to -\infty$), the variational principle of entropy maximization fundamentally transitions into a pure kinetic energy maximization problem. This transition forces the system to break the continuous zonal symmetry, breaking into concentration of absolute vorticity in the form of isolated point vortices or dipoles. While the existence and formation of these non-zonal structures are mathematically justified by the energy bounds, a critical physical question remains: once formed, are these highly concentrated dipole configurations dynamically stable under the evolution of the Euler equations on the rotating sphere?

In this chapter, we address the stability of the emergent dipole. We will first review the classical variational approach to hydrodynamical stability, specifically the Arnold stability criteria derived for spherical flows. Subsequently, we will show that while the dipole evades classical local stability bounds, it achieves robust non-linear stability through topological trapping as a global energy maximizer on its isovortical orbit.

5.1 Stability of the Emergent Dipole

To understand the dynamic stability of stationary solutions to the Euler equations, we utilize the energy-Casimir variational method originally developed by Arnold. The fundamental idea is to construct a generalized energy functional $\mathcal{H}[\omega]$ that is strictly conserved along the fluid trajectories, defined as the sum of the kinetic energy E and a generalized Casimir invariant C:

$$\mathcal{H}[\omega] = E[\omega] + C[q] = -\frac{1}{2}\int_{S^2} \psi\omega d\sigma + \int_{S^2} \Phi(q) d\sigma \tag{5.1}$$

where $q = \omega + f(\theta) = \omega + 2\Omega \sin\theta$ is the absolute vorticity, and Φ is an arbitrary smooth function.

5.1.1 First Variation and Stationary States

We consider a dynamically accessible perturbation to the relative vorticity field, $\omega \to \omega + \delta\omega$. Since the planetary vorticity $f(\theta)$ is strictly determined by the spherical geometry, the perturbation of the absolute vorticity is identical to that of the relative vorticity, namely $\delta q = \delta\omega$. The stream function also perturbs linearly as $\psi \to \psi + \delta\psi$, satisfying the Poisson relation $\Delta(\delta\psi) = \delta\omega$.

We compute the first variational derivative of the kinetic energy E. Expanding the perturbed energy and neglecting the second-order term $\mathcal{O}(\delta\psi\delta\omega)$ yields:

$$\delta E = -\frac{1}{2}\int_{S^2} (\psi\delta\omega + \delta\psi \cdot \omega) d\sigma \tag{5.2}$$

Exploiting the self-adjointness of the Laplace-Beltrami operator Δ on the sphere without boundaries, the second term can be rewritten via integration by parts:

$$\int_{S^2} \delta\psi \cdot \omega d\sigma = \int_{S^2} \delta\psi \Delta\psi d\sigma = \int_{S^2} \Delta(\delta\psi)\psi d\sigma = \int_{S^2} \delta\omega \cdot \psi d\sigma \tag{5.3}$$

Substituting this back, the factors of $1/2$ perfectly cancel out, giving the functional derivative of the kinetic energy:

$$\delta E = -\int_{S^2} \psi\delta\omega d\sigma. \tag{5.4}$$

Similarly, the first variation of the Casimir invariant is simply given by the chain rule:

$$\delta C = \int_{S^2} \Phi'(q)\delta q d\sigma = \int_{S^2} \Phi'(q)\delta\omega d\sigma \tag{5.5}$$

For a state to be a stationary point (equilibrium) of the Euler dynamics, the first variation of the generalized Hamiltonian must vanish for any arbitrary perturbation $\delta\omega$:

$$\delta\mathcal{H} = \int_{S^2} \left(-\psi + \Phi'(q)\right) \delta\omega d\sigma = 0 \implies \psi = \Phi'(q) \tag{5.6}$$

Assuming Φ' is invertible, this implies that the absolute vorticity must be a functional of the stream function, $q = (\Phi')^{-1}(\psi) := F(\psi)$. This recovers exactly the steady-state Euler equation $\Delta\psi + f(\theta) = F(\psi)$ discussed in Chapter 3. Here we showed that variational solution of $\mathcal{H}$ under constraints of Casimir invariants and energy conservation is a steady state of Euler's equation, whereas in Chapter 3 we showed that entropy maximizers correspond to a steady state of Euler's equation.

5.1.2 Second Variation and Arnold Stability for $F'(\psi) > -6$

To ascertain whether this stationary point is a local extremum (and hence Lyapunov stable) or a saddle point, we evaluate the second variation $\delta^2\mathcal{H}$. By differentiating $\delta\mathcal{H}$ again, utilizing $|\nabla\delta\psi|^2 = -\delta\psi\Delta(\delta\psi) = -\delta\psi\delta\omega$ and noting that $\Phi''(q) = \frac{1}{F'(\psi)}$, we obtain that

$$\delta^2 E = -\int_{S^2} \delta\psi\, \delta\omega\, d\sigma = -\int_{S^2} \delta\psi\, \Delta(\delta\psi)\, d\sigma = \int_{S^2} |\nabla\delta\psi|^2\, d\sigma$$

$$\delta^2 C = \int_{S^2} \Phi''(q)(\delta q)^2\, d\sigma = \int_{S^2} \Phi''(q)(\delta\omega)^2\, d\sigma$$

and therefore the quadratic form:

$$\delta^2\mathcal{H} = \int_{S^2} \left(|\nabla\delta\psi|^2 + \frac{1}{F'(\psi)}(\delta\omega)^2 \right) d\sigma \tag{5.7}$$

To analyse the sign of the quadratic form, we expand the perturbation in spherical harmonics. Since the perturbation preserves both the zero-mean condition and the angular momentum constraint, the modes corresponding to $l = 0$ and $l = 1$ are excluded. Hence we may write

$$\delta\psi = \sum_{l\geq 2} \sum_{m=-l}^{l} a_{l,m} Y_{l,m}$$

where the spherical harmonics satisfy $\Delta Y_{l,m} = -l(l+1)Y_{l,m}$. Therefore we have the equality,

$$\delta\omega = \Delta(\delta\psi) = -\sum_{l\geq 2}\sum_{m=-l}^{l} l(l+1)a_{l,m}Y_{l,m}$$

we obtain

$$\int_{S^2} |\nabla\delta\psi|^2\, d\sigma = \int_{S^2} \left\langle \sum_{l,m} a_{l,m}\,\nabla Y_{l,m}, \sum_{l',m'} a_{l',m'}\,\nabla Y_{l',m'} \right\rangle d\sigma = \sum_{l,m}\sum_{l',m'} a_{l,m}\overline{a_{l',m'}} \int_{S^2} \nabla Y_{l,m}\cdot\nabla\overline{Y_{l',m'}}\, d\sigma$$

and using the orthonormality of the spherical harmonics and Green's identity we have:

$$\int_{S^2} \nabla Y_{l,m}\cdot\overline{\nabla Y_{l',m'}}\, d\sigma = -\int_{S^2} Y_{l,m}\Delta\overline{Y_{l',m'}}\, d\sigma = l'(l'+1)\int_{S^2} Y_{l,m}\overline{Y_{l',m'}}\, d\sigma = l(l+1)\,\delta_{ll'}\delta_{mm'}$$

Plug in the above formula gives:

$$\int_{S^2} |\nabla\delta\psi|^2\, d\sigma = \sum_{l\geq 2}\sum_{m=-l}^{l} l(l+1)|a_{l,m}|^2$$

and

$$\int_{S^2} (\delta\omega)^2\, d\sigma = \sum_{l\geq 2}\sum_{m=-l}^{l} l^2(l+1)^2|a_{l,m}|^2$$

Substituting these into the above quadratic forms yields

$$\delta^2\mathcal{H} = \sum_{l\geq 2}\sum_{m=-l}^{l}\left[l(l+1) + \frac{l^2(l+1)^2}{F'(\psi)}\right]|a_{l,m}|^2$$

Equivalently,

$$\delta^2\mathcal{H} = \sum_{l\geq 2}\sum_{m=-l}^{l} l(l+1)\left(1 + \frac{l(l+1)}{F'(\psi)}\right)|a_{l,m}|^2$$

We now examine the sign of the coefficient. Since $l \geq 2$, we have $l(l+1) \geq 2(2+1) = 6$. If $F'(\psi) > 0$, then every coefficient is positive, so $\delta^2\mathcal{H}$ is positive definite. On the other hand, if $-6 < F'(\psi) < 0$ then for every $l \geq 2$,

$$1 + \frac{l(l+1)}{F'(\psi)} < 0$$

because $F'(\psi)$ is negative and $l(l+1) \geq 6 > -F'(\psi)$. Hence all coefficients are negative, so $\delta^2\mathcal{H}$ is negative definite. Therefore, the critical threshold is determined by the first nontrivial admissible mode $l = 2$, whose Laplace–Beltrami eigenvalue is 6.

According to the classical stability analysis for spherical domains, the sign of $F'(\psi)$ dictates the stability of the equilibrium:

- If $F'(\psi) > 0$ (Arnold's first theorem), both terms in the integrand are strictly positive. The quadratic form is positive-definite, indicating the stationary state is a strict local energy minimum and is therefore non-linearly stable.
- If $F'(\psi) < 0$, the second term introduces a negative contribution. However, Theorem 2 states that due to the discrete spectrum of the Laplace-Beltrami operator (with eigenvalues $0, -2, -6, \ldots$), the quadratic form remains negative-definite as long as $-6 < F'(\psi) < 0$. Hence the stationary state acts as a local energy maximum and retains Lyapunov stability.

5.1.3 Topological Trapping of the Dipole

We now apply this stability framework to the emergent dipole configuration derived in Chapter 4. For our variationally obtained steady state, the relationship between absolute vorticity and stream function is given by $F(\psi) = \lambda e^{\beta\psi}$, which yields:

$$F'(\psi) = \beta\lambda e^{\beta\psi} = \beta q. \tag{5.8}$$

In the negative temperature region ($\beta < 0$), since absolute vorticity $q > 0$, we have $F'(\psi) < 0$. However, there is a symmetry-breaking phase transition and a the highly concentrated dipole as the system is driven into the extreme limit where $\beta \to -\infty$. In this limit, the value of $F'(\psi)$ diverges to negative infinity and drops below the lower bound $F'(\psi) > -6$ established in Theorem 2.

This violation indicates that the local second-variation criteria developed for moderate zonal flows are insufficient to capture the stability of point-like singularities. Instead, the stability of the dipole must be understood macroscopically through the lens of global energy topology.

As established in Section 4.3, the localized vortex patch in the large negative β limit is not merely a local stationary point; it represents the absolute global maximum of the kinetic energy restricted to its corresponding isovortical orbit (a manifold defined by fixed Casimirs). Because the inviscid Euler equations strictly conserve both the total kinetic energy and all Casimir invariants, a fluid state initiated at this global maximum is topologically prohibited from evolving into any other configuration. Even in the presence of continuous finite perturbations, the system lacks the requisite energy budget to dislodge the dipole from the peak of this energy landscape. Consequently, the symmetry-breaking dipole, despite eluding local eigenvalue bounds, exhibits robust non-linear Lyapunov stability by virtue of being topologically trapped at the global variational extremum.

Rearrangement Inequality Define our rearrangement class (isovortical orbit) as:

$$\mathcal{O}_{q_0} = \{q_0 \circ \Phi^{-1} : \Phi \in \text{SDiff}(S^2)\}$$

Note that this can also be formulated using distribution functions as:

$$\mathcal{O}_{q_0} = \{q \in L^\infty(S^2) \mid \mu(\{x \in S^2 \mid q(x) > s\}) = \mu(\{x \in S^2 \mid q_0(x) > s\}) \quad \forall s \in \mathbb{R}\}$$

where μ is the Lebesgue measure on the sphere S^2. In this set $\mathcal{O}_{q_0}$, any vorticity field q is an area-preserving rearrangement of the initial field q_0. Since the Casimir invariants of the inviscid fluid are automatically satisfied on $\mathcal{O}_{q_0}$, we have successfully confined the dynamics in an infinite-dimensional space to a constrained manifold.

We define the absolute vorticity $\bar{q}$ that maximizes energy on the orbit:

$$\bar{q} \in \operatorname*{arg\,max}_{q \in \mathcal{O}_{q_0}} E(q)$$

Here the kinetic energy is expressed in terms of the stream function ψ, and is determined by

$$\Delta\psi = \omega = q - f$$

where $f(\theta) = 2\Omega \sin\theta$ denotes the Coriolis term. The energy functional therefore reads

$$E(q) = \frac{1}{2}\int_{S^2} |\nabla\psi|^2 \, d\sigma = -\frac{1}{2}\int_{S^2} \psi(q - f) \, d\sigma$$

This formulation highlights that the maximization is not performed over the entire function space, but rather over the dynamically invariant manifold $\mathcal{O}_{q_0}$, which represents the set of all vorticity rearrangements accessible under Euler evolution, and that energy $E(q(t)) = E(q_0)$ is conserved along the orbit.

Orbital Lyapunov Stability via Coercivity The extremal state $\bar{q}$ is not an isolated point, since the energy functional is invariant under rigid rotations of the sphere. If $R \in SO(3)$ denotes a rotation, then in case of no Coriolis force we have:

$$E(\bar{q} \circ R) = E(\bar{q})$$

Hence the maximizers form a symmetry orbit and we therefore seek stability modulo rotations.

$$\mathcal{M} = \{\bar{q} \circ R : R \in SO(3)\}$$

Here we remark that $M \subset \mathcal{O}_{q_0}$. Otherwise, the distance to M is not meaningful. If maximizer $\bar{q}$ itself belongs to $\mathcal{O}_{q_0}$; then its rotations also belong (since rotations are area-preserving diffeomorphisms) and thus:

$$R \in SO(3) \subset \text{SDiff}(S^2) \implies \bar{q} \circ R \in \mathcal{O}_{q_0}$$

Remark: in the following proposition we consider $\text{Stab}(f) \subset SO(3)$, the rotation group that leaves f invariant:

$$\text{Stab}(f) := \{R \in SO(3) : f \circ R^{-1} = f\}$$

here in this case with $f(\theta) = 2\Omega \sin\theta$, we have $\text{Stab}(f) \cong SO(2)$ (the rotations about the axis of rotation). Now define

$$\tilde{\mathcal{M}} := \{q \circ R : R \in \text{Stab}(f)\}$$

and the distance

$$d(q, \tilde{\mathcal{M}}) := \inf_{R \in SO(2)} \|q - \bar{q} \circ R\|_{H^{-1}}$$

then due to the subset relations $\tilde{\mathcal{M}} \subset \mathcal{M}$ we know that $d(q, \tilde{\mathcal{M}}) \geq d(q, \mathcal{M})$.

Definition (Local Orbital coercivity): We say that $\bar{q}$ is coercive on the orbit if there exist constants $c > 0$ and $\varepsilon > 0$ such that for all $q \in \mathcal{O}_{q_0}$ satisfying

$$\inf_{R \in SO(2)} \|q - \bar{q} \circ R\|_{H^{-1}} < \varepsilon$$

the following inequality holds:

$$E(\bar{q}) - E(q) \geq c \inf_{R \in SO(2)} \|q - \bar{q} \circ R\|_{H^{-1}}^2$$

This inequality quantifies the energetic penalty required to depart from the dipole configuration (modulo rigid rotations).

Lemma (Quadratic form of Energy) Fix any $\bar{q} \in H^{-1}(S^2)$. Then for any $q \in H^{-1}(S^2)$ it holds that:

$$|E(q) - E(\bar{q})| \leq C\|q - \bar{q}\|_{H^{-1}} \big(\|q - f\|_{H^{-1}} + \|\bar{q} - f\|_{H^{-1}}\big)$$

In particular, if $\|q - \bar{q}\|_{H^{-1}} \leq \delta$ and $\|q - f\|_{H^{-1}}, \|\bar{q} - f\|_{H^{-1}}$ are bounded by a constant M, then

$$|E(q) - E(\bar{q})| \leq C_M \|q - \bar{q}\|_{H^{-1}}$$

Moreover, if additionally $DE(\bar{q}) = 0$ along admissible directions (i.e. $\bar{q}$ is a constrained critical point), then locally we have the quadratic bound

$$E(\bar{q}) - E(q) \leq C_1 \|q - \bar{q}\|_{H^{-1}}^2$$

Proof. First define the operator $G := (-\Delta)^{-1}$, then we know $\psi = -G(q-f)$ and the energy could be written in terms of the quadratic form:

$$E(q) = \frac{1}{2}\int_{S^2} |\nabla\psi|^2 \, d\sigma = -\frac{1}{2}\int_{S^2} \psi(q-f)\, d\sigma = \frac{1}{2}\langle q-f, G(q-f)\rangle$$

Then by choosing the perturbation $q_\varepsilon = q + \varepsilon h$. A direct expansion gives:

$$E(q+\varepsilon h) = \frac{1}{2}\langle q-f+\varepsilon h, G(q-f+\varepsilon h)\rangle = \frac{1}{2}\langle q-f, G(q-f)\rangle + \varepsilon\langle h, G(q-f)\rangle + \frac{\varepsilon^2}{2}\langle h, Gh\rangle$$

where the first variation (Frechet derivative) is $DE(q)[h] = \langle h, G(q-f)\rangle$ or equivalently:

$$\frac{\delta E}{\delta q} = G(q-f) = -\psi$$

The second variation is a constant bilinear form $D^2E(q)[h,h] = \langle h, Gh\rangle$ which is independent of q. Moreover, using the operator bound $\|Gh\|_{H^1} \le C\|h\|_{H^{-1}}$ we obtain that:

$$|\langle h, Gh\rangle| \le \|h\|_{H^{-1}}\|Gh\|_{H^1} \le C\|h\|^2_{H^{-1}}$$

Therefore we conclude the Hessian is bounded, but how are we going to relate back to the energy difference?

Now let $h := q - \bar{q}$ then applying the exact expansion above to $q = \bar{q}$ with perturbation h:

$$E(\bar{q}+h) = E(\bar{q}) + DE(\bar{q})[h] + \frac{1}{2}\langle h, Gh\rangle$$

Rearranging this becomes:

$$E(q)-E(\bar{q}) = DE(\bar{q})[h]+\frac{1}{2}\langle h, Gh\rangle = \langle h, G(\bar{q}-f)\rangle+\frac{1}{2}\langle h, G((q-f)-(\bar{q}-f))\rangle = \frac{1}{2}\langle h, G(q-f)\rangle+\frac{1}{2}\langle h, G(\bar{q}-f)\rangle$$

Written in short this is

$$E(q) - E(\bar{q}) = \frac{1}{2}\Big\langle q-\bar{q},\ G\big((q-f)+(\bar{q}-f)\big)\Big\rangle$$

Taking absolute values and applying the dual pairing estimate gives

$$|E(q) - E(\bar{q})| \le \frac{1}{2}\|q-\bar{q}\|_{H^{-1}} \left\|G\big((q-f)+(\bar{q}-f)\big)\right\|_{H^1}$$

Now use the elliptic estimate for $G = (-\Delta)^{-1}$, where $\|Gu\|_{H^1} \le C\|u\|_{H^{-1}}$ we obtain

$$|E(q) - E(\bar{q})| \le C\|q-\bar{q}\|_{H^{-1}} \|(q-f)+(\bar{q}-f)\|_{H^{-1}}$$

Consequently,

$$|E(q) - E(\bar{q})| \le C\|q-\bar{q}\|_{H^{-1}}\Big(\|q-f\|_{H^{-1}} + \|\bar{q}-f\|_{H^{-1}}\Big)$$

In particular, if $\|q-f\|_{H^{-1}},\ \|\bar{q}-f\|_{H^{-1}} \le M$ are bounded then we have

$$|E(q) - E(\bar{q})| \le C_M\|q-\bar{q}\|_{H^{-1}}$$

Moreover, if $h := q - \bar{q}$ satisfies $DE(\bar{q})[h] = 0$, then $E(q) - E(\bar{q}) = \frac{1}{2}\langle h, Gh\rangle$ and therefore using again the dual pairing estimate together with the elliptic bound:

$$|E(q) - E(\bar{q})| \le C\|q-\bar{q}\|^2_{H^{-1}}$$

which completes the proof. □

Proposition (Orbital Lyapunov stability): Assume the coercivity estimate holds. Let $q(t)$ be a solution of the Euler equation with initial data $q(0) = q_0$ sufficiently close to $\tilde{\mathcal{M}}$ under H^{-1} norm, then we have:

$$\sup_{t\geq 0} d(q(t), \tilde{\mathcal{M}}) \leq C d(q_0, \tilde{\mathcal{M}}), \quad \text{where} \quad d(q, \tilde{\mathcal{M}}) := \inf_{R\in SO(2)} \|q(t) - \bar{q}\circ R\|_{H^{-1}}$$

Proof. Define the Lyapunov functional $V(q) = E(\bar{q}) - E(q)$. Since energy is conserved along Euler trajectories, we therefore have:

$$V(q(t)) = V(q_0)$$

Applying coercivity yields

$$\inf_{R\in SO(2)} \|q(t) - \bar{q}\circ R\|^2_{H^{-1}} \leq \frac{1}{c} V(q) = \frac{1}{c} V(q_0)$$

Since the lemma gives an estimate between $\bar{q}$ and $q(t)$, evaluate at $t = 0$ where $q(0) = q_0$ yields:

$$V(q_0) = E(\bar{q}) - E(q_0) \leq C_1 \|q_0 - \bar{q}\|^2_{H^{-1}}$$

and since the lemma is true for any reference $\bar{q}$, by assigning $\bar{q} \leftarrow \bar{q}\circ R$ and taking the infimum over all rotational symmetry subgroup $\tilde{\mathcal{M}}$ we have:

$$E(\bar{q}\circ R) - E(q_0) \leq C_1 \inf_{R\in SO(2)} \|q_0 - \bar{q}\circ R\|^2_{H^{-1}}$$

But we know that energy is conserved under rotations in $SO(2)$, therefore we have:

$$V(q_0) = E(\bar{q}) - E(q_0) = E(\bar{q}\circ R) - E(q_0) \leq C_1 \inf_{R\in SO(2)} \|q_0 - \bar{q}\circ R\|^2_{H^{-1}}$$

and that

$$\inf_{R\in SO(2)} \|q(t) - \bar{q}\circ R\|_{H^{-1}} \leq \frac{\sqrt{C_1}}{\sqrt{c}} \inf_{R\in SO(2)} \|q_0 - \bar{q}\circ R\|_{H^{-1}}$$

and finally by redefine the constant we arrive at:

$$\sup_{t\geq 0} \inf_{R\in SO(2)} \|q(t) - \bar{q}\circ R\|_{H^{-1}} \leq C \inf_{R\in SO(2)} \|q_0 - \bar{q}\circ R\|_{H^{-1}} = C d(q_0, \tilde{M})$$

which returns our desired inequality:

$$\sup_{t\geq 0} d(q(t), \tilde{\mathcal{M}}) \leq C d(q_0, \tilde{\mathcal{M}})$$

□

Interpretation of Topological Trapping Let q(t) be a solution of the Euler equation with initial data q_0 sufficiently close to the dipole orbit $\tilde{\mathcal{M}}$ in H^{-1}. Then the trajectory remains confined in a tubular neighbourhood of $\tilde{\mathcal{M}}$ for all time. This shows that the dipole configuration acts as a topologically trapped state: perturbations cannot drift away from the dipole orbit under the Euler dynamics.

The above result shows that the dipole maximizer is orbitally stable modulo the rotational symmetry SO(2). In particular, small perturbations cannot escape the dipole configuration but remain confined to a neighbourhood of the rotation orbit. This provides a rigorous mathematical justification for the long-time persistence of dipole structures observed in two-dimensional turbulence on the rotating sphere.

Chapter 6

Conclusion

Appendix A

Required Mathematical Tools and Physics Concepts

A.0.1 Poisson Brackets and Hamiltonian

We begin by introducing Poisson Brackets on the plane $\mathbb{R}^2$. Let $f(x, y)$ and $g(x, y)$ be two smooth functions. Define the **Poisson bracket** as:

$$\{f, g\} = \partial_x f \, \partial_y g - \partial_y f \, \partial_x g$$

which can be viewed as a 2×2 determinant:

$$\{f, g\} = \det \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}$$

Then by defining the rotated gradient $\nabla^\perp f = (-f_y, f_x)$, a direct computation gives:

$$\nabla^\perp f \cdot \nabla g = (-f_y, f_x) \cdot (g_x, g_y) = -f_y g_x + f_x g_y = \{f, g\}$$

Recall the definition of directional derivative, the poisson brackets then measures the change of g along level sets of f, since $\nabla^\perp f$ is perpendicular to ∇f and therefore tangent to the level curves.

In classical mechanics, a Hamiltonian system is formulated in phase space coordinates (q, p), representing position and momentum respectively. Given the Hamiltonian $H(q, p)$, with dynamics:

$$\dot{q} = -\partial_p H, \quad \dot{p} = \partial_q H$$

For clarity reasons, we now suppress mechanical meanings of these variables, rename them as simply (x, y) on the phase space and consider $H(x, y)$. We sometimes use the matrix J to represent rotation of the gradients. Then if $\mathbf{z} = (x, y)$ we have:

$$\dot{\mathbf{z}} = J \nabla H = (-H_y, H_x) = \nabla^\perp H, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

For our two-dimensional inviscid incompressible flow, with $\mathbf{u}$ is the velocity field and ω the vorticity. We have:

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = 0$$

Then the incompressible condition $\nabla \cdot \mathbf{u} = 0$ implies there exists a stream function ψ such that $\mathbf{u} = \nabla^{\perp}\psi = J\nabla\psi$. Moreover, the vorticity and stream function is linked by:

$$\omega = \operatorname{curl} \mathbf{u} = \Delta\psi$$

Now we consider particle trajectories associated with the velocity field. For each initial point $x_0 \in \Omega$, define the trajectory $X(t; x_0)$ by:

$$\frac{d}{dt}X(t; x_0) = \mathbf{u}(t, X(t; x_0)), \quad X(0; x_0) = x_0$$

Therefore particle trajectories satisfy:

$$\frac{d}{dt}X(t; x_0) = \mathbf{u}(t, X(t; x_0)) = J\nabla\psi(t, X(t; x_0))$$

Thus the particle motion is a Hamiltonian system with time-dependent Hamiltonian H given by the stream function ψ. We remark that upon substitution of $\mathbf{u} = \nabla^{\perp}\psi$, we are able to rewrite our Euler's equation in Poisson Brackets Formulations:

$$\partial_t\omega + \{\psi, \omega\} = 0$$

A.0.2 Casimir Invariants

A Casimir invariant is a functional $C[\omega]$ satisfying $\{C, F\} = 0$ for all admissible functionals F. The standard Poisson Brackets for finite canonical system is given by:

$$\{F, G\} = \sum_{i=1}^{n}(\partial_{q_i}F\partial_{p_i}G - \partial_{p_i}F\partial_{q_i}G)$$

which is non-degenerate, meaning that $\{C, G\} = 0 \quad \forall G \implies C =$ constant. Therefore in the finite canonical structure there are almost no nontrivial Casimirs. However, for infinite dimensional vorticity formulation, the Poisson bracket is defined on the functionals:

$$\{F, G\}(\omega) = \int \omega \left\{\frac{\delta F}{\delta\omega}, \frac{\delta G}{\delta\omega}\right\} dx$$

Here the bracket on the right-hand side is the spatial Poisson bracket.

First consider an unbounded/periodic planar domain: the 2D Euler vorticity equation (or on a rotating sphere using absolute vorticity q, the logic is the same)

$$\partial_t\omega + \mathbf{u} \cdot \nabla\omega = 0, \quad \nabla \cdot \mathbf{u} = 0$$

Take any $f \in C^1$. By the chain rule:

$$\partial_t f(\omega) = f'(\omega)\partial_t\omega, \quad \nabla f(\omega) = f'(\omega)\nabla\omega$$

Multiplying the PDE by $f'(\omega)$ (equivalently, composing ω with f):

$$\partial_t f(\omega) + \mathbf{u} \cdot \nabla f(\omega) = 0$$

Now using the Divergence Theorem with no boundary gives:

$$u \cdot \nabla f(\omega) = \nabla \cdot (\mathbf{u} f(\omega)) - f(\omega) \nabla \cdot \mathbf{u} = \nabla \cdot (\mathbf{u} f(\omega))$$

$$\partial_t f(\omega) + \nabla \cdot (\mathbf{u} f(\omega)) = 0$$

Integrating over the entire domain (no boundary, periodic boundary, or impermeable boundary satisfying $\mathbf{u} \cdot \mathbf{n} = 0$), and applying the divergence theorem:

$$\frac{d}{dt} \int f(\omega)\, dx = -\int \nabla \cdot (\mathbf{u} f(\omega))\, dx = -\int_{\partial\Omega} f(\omega) \mathbf{u} \cdot \mathbf{n}\, dS = 0.$$

Therefore $\int f(\omega)\, dx$ is conserved. The same holds on the sphere: replace dx with the area element $d\sigma$, and $\nabla\cdot$ with the spherical divergence. Since the sphere has no boundary, the boundary flux term automatically vanishes. Thus all functionals of the form

$$\int_{S^2} f(q)\, d\sigma$$

are conserved. (Here we have to consider the absolute vorticity q on a sphere)

A.0.3 Spherical Harmonics

Let $S^2 = \{x \in \mathbb{R}^3 : |x| = 1\}$ be the unit sphere equipped with the surface measure $d\sigma$. The Laplace–Beltrami operator Δ_{S^2} admits a complete orthonormal system of eigenfunctions

$$\{Y_{l,m}\}_{l \geq 0,\, |m| \leq l} \subset C^\infty(S^2),$$

called spherical harmonics, satisfying

$$\Delta_{S^2} Y_{l,m} = -l(l+1) Y_{l,m}.$$

The eigenspace $H_l := \operatorname{span}\{Y_{l,m}\}_{|m| \leq l}$ has dimension $2l + 1$, and we have the orthogonal decomposition

$$L^2(S^2) = \bigoplus_{l=0}^{\infty} H_l, \quad f = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} c_{l,m} Y_{l,m}, \quad c_{l,m} = \int_{S^2} f \overline{Y_{l,m}}\, d\sigma.$$

A.0.4 Self-Propagation of a Dipole Configuration on S^2

In the following, $x, y \in S^2$ always denote points on the unit sphere. The Green's function $G : S^2 \times S^2 \to \mathbb{R}$ of the Laplace–Beltrami operator is defined (with a zero-mean normalisation) by

$$\Delta_x G(x, y) = \delta_y(x) - \frac{1}{4\pi}, \qquad \int_{S^2} G(x, y)\, d\sigma_x = 0, \tag{A.1}$$

where Δ_x indicates the Laplacian acting on the *first* variable x. Consequently, for any vorticity field ω with zero mean, the stream function satisfies $\Delta\psi = \omega$ and admits the Green representation

$$\psi(x) = \int_{S^2} G(x, y)\, \omega(y)\, d\sigma_y. \tag{A.2}$$

Here the stream function $\psi(x)$ still satisfies $\Delta\psi = \omega$ after normalization by $1/4\pi$ and we provide a sanity check by:

$$\Delta_x \psi(x) = \int_{S^2} \delta_y(x) \omega(y)\, d\sigma_y - \frac{1}{4\pi} \int_{S^2} \omega(y)\, d\sigma_y = \omega(x)$$

Dipole ansatz. Consider a (formal) dipole configuration

$$\omega(x,t) = \Gamma\,\delta_{x_+(t)}(x) - \Gamma\,\delta_{x_-(t)}(x), \qquad \Gamma > 0 \tag{A.3}$$

Plugging (A.3) into (A.2) yields

$$\psi(x,t) = \Gamma G(x, x_+(t)) - \Gamma G(x, x_-(t)) \tag{A.4}$$

Velocity field. The incompressible velocity is given by $u(x,t) = \nabla_x^\perp \psi(x,t)$, where ∇_x denotes the surface gradient with respect to the first variable x (so y is treated as a parameter), and $\nabla_x^\perp := J\,\nabla_x$ is the $90°$ rotation in the tangent plane (with J the usual complex structure on TS^2). From (A.4) we obtain

$$u(x,t) = \Gamma \nabla_x^\perp G(x, x_+(t)) - \Gamma \nabla_x^\perp G(x, x_-(t)) \tag{A.5}$$

Point-vortex transport law. Formally, each point vortex is transported by the velocity field generated by the *other* vortex:

$$\dot{x}_+(t) = u(x_+(t), t), \qquad \dot{x}_-(t) = u(x_-(t), t) \tag{A.6}$$

where the self-induced term is discarded/renormalised (self-interaction). Using (A.5) gives

$$\dot{x}_+(t) = -\Gamma \nabla_1^\perp G(x_+(t), x_-(t)), \qquad \dot{x}_-(t) = \Gamma \nabla_1^\perp G(x_-(t), x_+(t)) \tag{A.7}$$

Key symmetry identity. We now prove the crucial relation

$$\nabla_1^\perp G(x,y) = -\nabla_1^\perp G(y,x) \quad \text{for } x, y \in S^2, x \neq y \tag{A.8}$$

Step A: Symmetry and the swap of arguments. The Green's function is symmetric, $G(x,y) = G(y,x)$. Let ∇_1 and ∇_2 denote the gradients with respect to the first and second argument slots, respectively. Differentiating the symmetry identity $G(x,y) = G(y,x)$ with respect to x yields:

$$\nabla_1 G(x,y) = \nabla_2 G(y,x) \tag{A.10}$$

where on the right-hand side, since x appears in the second slot, the derivative is taken with respect to that slot and evaluated at (y,x).

Step B: Isotropy and anti-symmetry of gradients. Since the Laplace-Beltrami operator is invariant under rotations, $G(x,y)$ depends only on the geodesic distance $d(x,y)$. Thus, $G(x,y) = g(d(x,y))$. By the chain rule:

$$\nabla_1 G(x,y) = g'(d)\nabla_x d(x,y), \quad \nabla_2 G(x,y) = g'(d)\nabla_y d(x,y) \tag{A.11}$$

Using the geometric identity $\nabla_x d(x,y) = -\nabla_y d(x,y)$ (unit vectors pointing in opposite directions along the geodesic), we obtain:

$$\nabla_1 G(x,y) = -\nabla_2 G(x,y) \tag{A.12}$$

Step C: Synthesis and self-propagation. Combining (A.10) and (A.12), we have:

$$\nabla_1 G(x,y) = \nabla_2 G(y,x) = -\nabla_1 G(y,x) \tag{A.13}$$

By viewing S^2 as embedded in $\mathbb{R}^3$, the rotation operator J can be identified with the cross product $J_p v = p \times v$ for $v \in T_p S^2$. Since both sides of (A.13) are equal as vectors in $\mathbb{R}^3$, applying the rotation $v \mapsto p \times v$ to each respective side yields:

$$\nabla_1^{\perp} G(x, y) = -\nabla_1^{\perp} G(y, x) \tag{A.14}$$

This confirms that in the ODE system (A.6), the velocities $\dot{x}_+$ and $\dot{x}_-$ satisfy:

$$\dot{x}_+(t) = -\Gamma \nabla_1^{\perp} G(x_+, x_-) = \Gamma \nabla_1^{\perp} G(x_-, x_+) = \dot{x}_-(t) \tag{A.15}$$

Therefore the two vortices retain a fixed separation while translating together along a great circle. This rigid translation of the dipole is precisely the *self-propagation* phenomenon: each vortex is advected by the velocity field generated by the other, and the symmetry of the Green kernel forces the two advection velocities to coincide.

Appendix B

Bibliography

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